

PROFESSIONAL PROFILE

Lead Data and MLOps Engineer specializing in production-grade generative AI, machine learning, and data platforms. Designs secure evaluation, monitoring, inference, and analytics solutions across AWS, Snowflake, and Apache Airflow, with a strong record of partnering with data scientists and business stakeholders to operationalize reliable, governed AI systems.

GENERATIVE AI

Evaluation, agents, enterprise analytics

MLOPS PLATFORMS

Monitoring, inference, model operations

CLOUD & DATA

AWS, Snowflake, Airflow, Terraform

PROFESSIONAL EXPERIENCE

Lead Data Engineer (MLOps) | Zelis

Remote | Aug 2021 - Present

Lead MLOps and data engineering initiatives that operationalize generative AI and traditional machine learning solutions across AWS, Snowflake, and Apache Airflow.

SELECTED AI/ML PLATFORMS

- **GenAI Evaluation Platform:** Developed an enterprise platform for evaluating company GenAI applications and custom GPTs, including a reusable Python metrics library, AWS Step Functions ETL workflows, and an AWS-hosted Streamlit dashboard with SSO and role-based access control.
- **Model Monitoring Platform:** Designed daily data drift and model performance monitoring with Airflow, AWS Lambda, Amazon ECS, a custom Snowflake data model, Streamlit reporting, and automated email alerts for critical model issues.
- **Batch Inference Pipelines:** Built Airflow-orchestrated SageMaker and SageMaker Pipelines workflows for traditional ML inference, persisting prediction outputs to Snowflake for reporting and downstream analysis.
- **Agentic Jira Assistant:** Created a Streamlit application powered by a custom AI agent that uses project context and brief user requests to generate structured Jira descriptions, acceptance criteria, and implementation details.
- **ChatGPT Logging & Secure Explorer:** Implemented daily Amazon ECS jobs using the OpenAI Compliance API to retrieve Enterprise conversations, store standardized JSON in Amazon S3, generate custom GPT datasets, and provide authorized users an authenticated search, view, and download interface.

ENGINEERING ENABLEMENT & INTERNAL PLATFORMS

- **Private PyPI Repository:** Established an AWS-hosted package index for distributing internal Python source and wheel packages through pip within the corporate network.
- **ChatGPT Usage Dashboard:** Built an AWS-hosted Streamlit application that reports OpenAI Enterprise adoption, usage patterns, and business-purpose metrics.
- **Team Deployment Portal:** Developed a Flask application integrated with GitHub Enterprise and GitHub Actions to trigger environment-specific deployments, persist user and workflow metadata, and display asynchronous deployment status.
- **Team Documentation Portal:** Created a Flask knowledge hub for applications, processes, databases, and SOPs, with executive-friendly technical content, interactive Mermaid diagrams, and cross-category tag filtering.

PLATFORM ENGINEERING, GOVERNANCE & DELIVERY

- **Security & Access Governance:** Implemented SSO, role-based access control, and restrictive AWS permissions to protect sensitive enterprise AI data and internal applications.
- **Cloud Infrastructure & Delivery:** Automated repeatable application and platform deployments with Terraform, Docker, Amazon ECR, ECS, Lambda, IAM, SSO, and CI/CD workflows.
- **Large-Scale Data Processing:** Developed Spark and Amazon EMR pipelines to process high-volume datasets and support scalable analytics and machine learning workflows.
- **Snowflake & Airflow Engineering:** Designed database structures and production data pipelines that integrate Snowflake with Apache Airflow for dependable, repeatable processing.
- **Cross-Functional Delivery:** Partnered with data scientists, engineers, and business stakeholders to productionize models and deliver secure, maintainable AI/ML systems.

Senior Data Analyst | C.H. Robinson

Remote | Jan 2014 - Aug 2021

- Conducted exploratory analysis in Jupyter to identify trends and deliver actionable business insights.
- Built interactive analytics applications and dashboards with Plotly, Streamlit, Power BI, and R Shiny.
- Optimized SQL queries and data workflows across SQL Server, PostgreSQL, and Hive environments.
- Developed Python-based ETL, data integration, and automated reporting solutions.
- Improved data quality by integrating and validating disparate data sources and translating stakeholder requirements into scalable solutions.

TECHNICAL SKILLS

PROGRAMMING & DATA ENGINEERING Python, SQL, Apache Airflow, Spark, Docker, Bash, ETL/ELT, API integration	GENERATIVE AI & MLOPS GenAI evaluation, AI agents, model monitoring, data drift, batch inference, usage analytics
AWS SageMaker, Step Functions, ECS, ECR, Lambda, S3, IAM, RDS, Glue, EMR	DATA PLATFORMS Snowflake, PostgreSQL, SQL Server, Hive, database design, data modeling
APPLICATIONS & VISUALIZATION Streamlit, Flask, Jupyter, Plotly, Power BI, R Shiny, Mermaid	INFRASTRUCTURE & CI/CD Terraform, GitHub Actions, GitHub Enterprise, SSO, role-based access control

AI-AUGMENTED ENGINEERING PRACTICES

- **AI Pair Programming:** Use Cursor, GitHub Copilot, and Codex to accelerate test generation, boilerplate, refactoring support, and routine engineering tasks while maintaining review ownership of production code.
- **Agent-Assisted Documentation:** Direct coding agents to draft technical documentation, summarize implementation details, and explain completed work to improve maintainability, handoffs, and team knowledge sharing.

EDUCATION

University of Massachusetts Lowell
B.S. Information Technology
Focus: programming and relational databases

University of Oklahoma
Programming and relational database coursework
Research assistant, NSSL | Minor in Mathematics